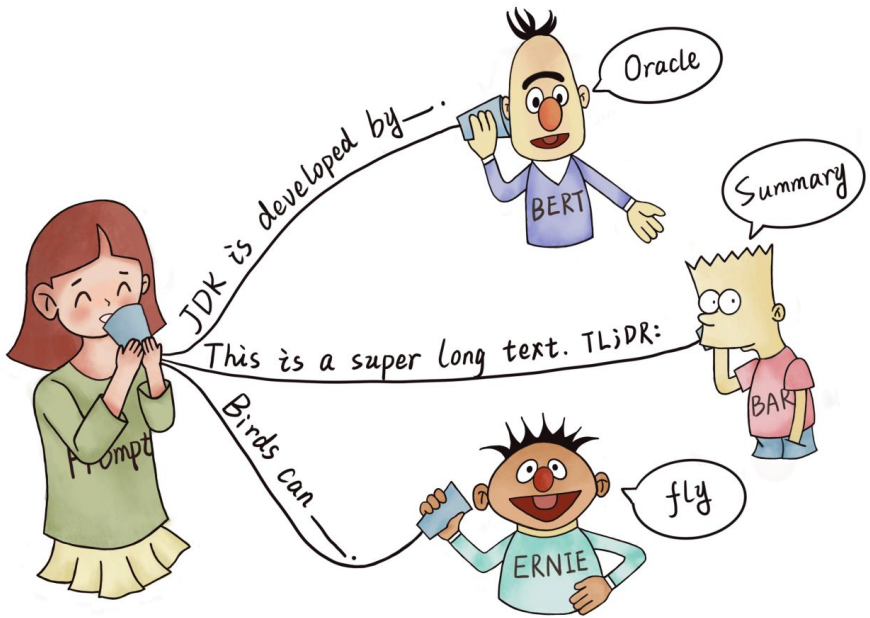
INFO-I590 Fundamentals and Applications of LLMs

Prompting

Jisun An

What is Prompting?

- Encouraging a pre-trained model to make particular predictions by providing a textual “prompt” specifying the task to be done



img: https://arxiv.org/abs/2107.13586

Decoding / Inference

Choosing the Next Token in Generation

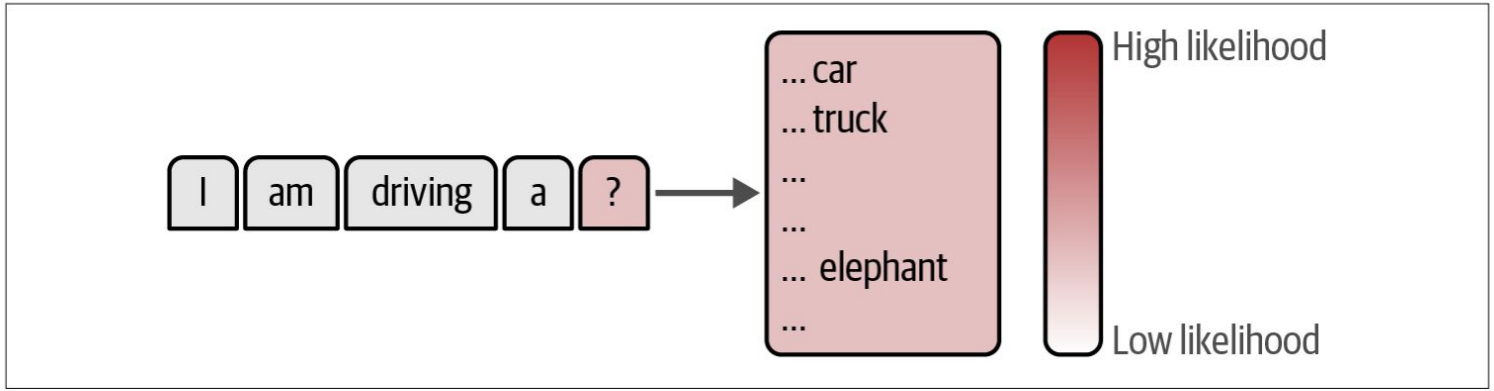
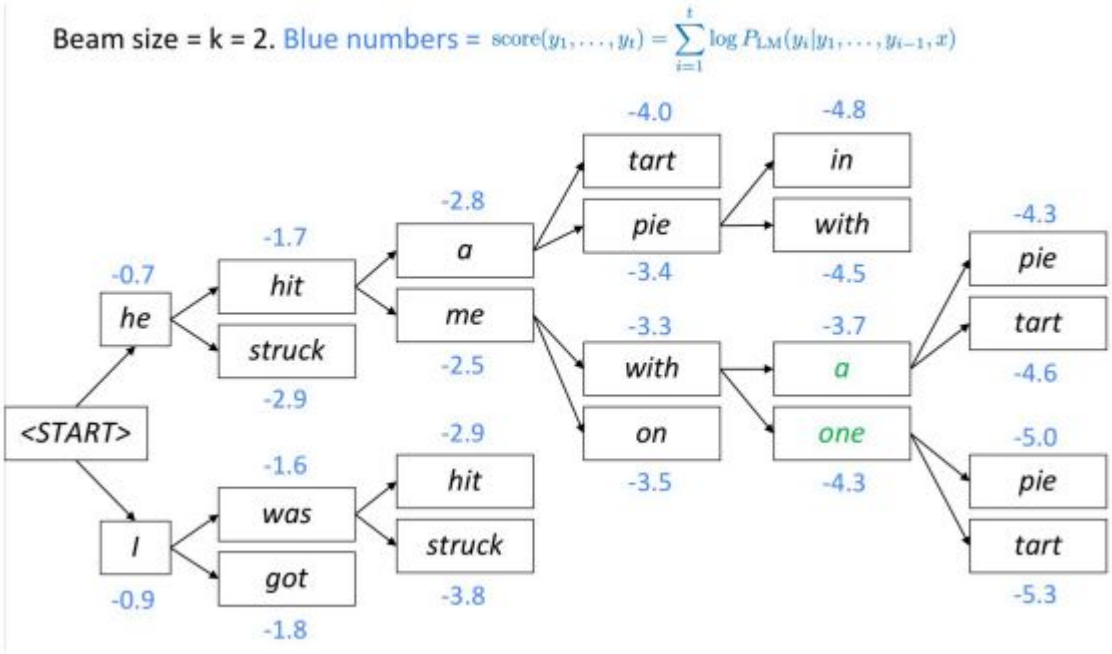


Figure 6-3. The model chooses the next token to generate based on their likelihood scores.

Decoding algorithms

- **Greedy decoding** is a simple method; gives low quality output
  - Select the most probable word (argmax)
- **Sampling methods** are a way to get more diversity and randomness
  - **Pure sampling**: On each step t, randomly sample from the probability distribution P_t to obtain your next word.
  - **Top-n sampling**: On each step t, randomly sample from P_t, restricted to just the **top-n most probable words**
  - Good for open-ended / creative generation (poetry, stories)
- **Beam search** explores several different hypotheses instead of just a single one
  - keep track of *k* most probable partial translations at each decoder step
  - the beam size k is usually 5-10

Beam Search (k=2) Example



img: https://atcold.github.io/NYU-DLSP20/en/week12/12-2/

Prompting Fundamentals

Basic Prompting (Radford et al. 2018)

- Append a textual string to the beginning of the sequence and complete

x = **When a dog sees a squirrel, it will usually**

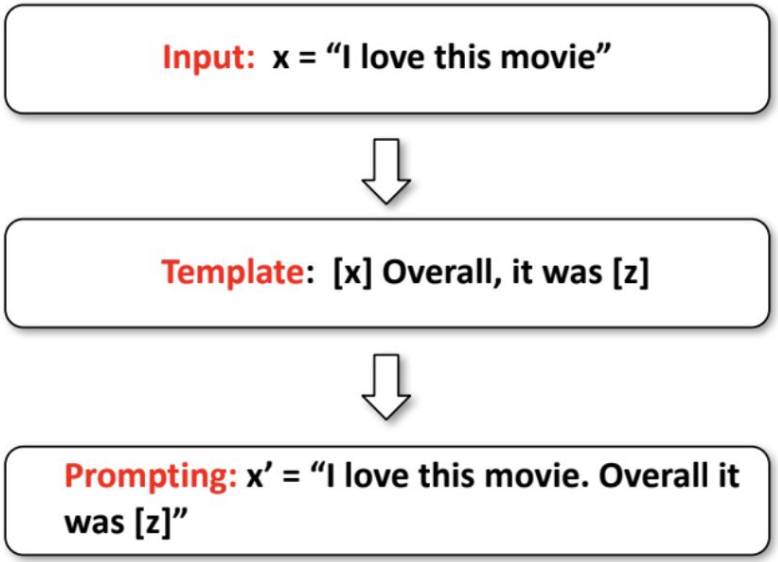
(GPT-2 Small) be afraid of anything unusual. As an exception, that's when a squirrel is usually afraid to bit.
(GPT-2 XL) lick the squirrel. It will also touch its nose to the squirrel on the tail and nose if it can.

Standard Prompting Workflow

- Fill a prompt template
- Predict the answer
- Post-process the answer

Prompt Templates

- A template where you fill in with an actual input



Chat Prompts

- Recently, many models are trained as chatbots
- Usually inputs are specified in OpenAI messages format

```
messages=[
  {
    "role": “system",
    "content": “Please classify movie reviews as 'positive' or ‘negative'."
  },
  {
    "role": “user",
    "content": "This movie is a banger.”
  },
]
```

- Roles
  - “system”: message provided to the system to influence behavior
  - “user”: message input by the user
  - “assistant”: message output by the system

Chat Prompts Behind the Scenes

- Behind the scenes, messages are converted to token strings

LLaMa

Sys.
```
[INST]
<<SYS>>
You are an assistant that …
<</SYS>>
[/INST]
```

User
```
[INST]This movie is great.[/INST]
```

Asst.
```
Positive.
```

Alpaca

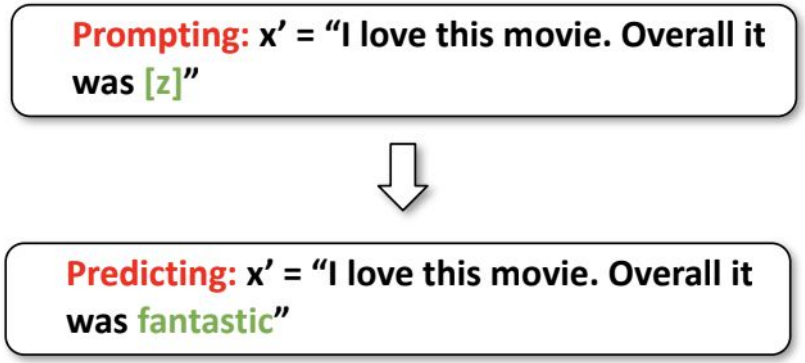
Sys.
```
### Instruction:
You are an assistant that …
```

User
```
### Instruction:
This movie is great.
```

Asst.
```
### Response:
Positive.
```

Answer Prediction

- Given a prompt, predict the answer



- Use any inference algorithms (sampling, greedy method, etc)

Post-processing

- Based on the answer, select the actual output
- For instance:
  - Taking the output as-is
  - Formatting the output for easy visualization
  - Selecting only parts of the output that you want to use
  - Mapping the outputs to other actions

Output Formatting

Markdown Rendering

You
Please write a table with the last 5 presidents and their birthdates.

ChatGPT
Here is a table with the last 5 U.S. Presidents and their birthdates:

President	Birthdate
Joe Biden	November 20, 1942
Donald Trump	June 14, 1946
Barack Obama	August 4, 1961
George W. Bush	July 6, 1946
Bill Clinton	August 19, 1946

Code

You
Output that in JSON format.

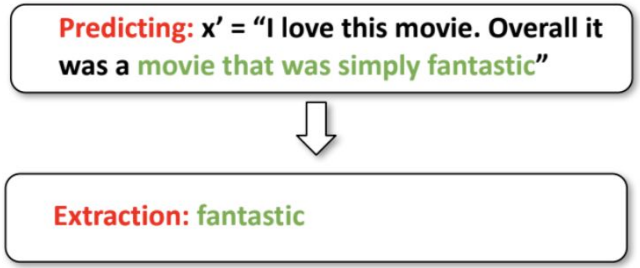
ChatGPT
Here is the information in JSON format:

```
[
  {"President":"Joe Biden","Birthdate":"November 20, 1942"},
  {"President":"Donald Trump","Birthdate":"June 14, 1946"},
  {"President":"Barack Obama","Birthdate":"August 4, 1961"},
  {"President":"George W. Bush","Birthdate":"July 6, 1946"},
  {"President":"Bill Clinton","Birthdate":"August 19, 1946"}
]
```
```​`` 【oaicite:0】 ``​
```

- For user-facing applications, format in a pretty way

Output Selection

- From a longer response, select the information indicative of an answer

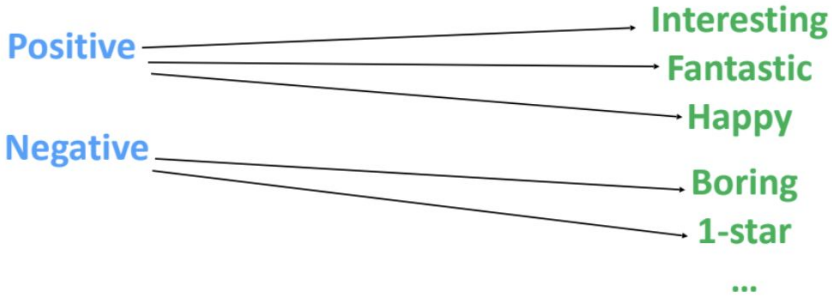


- Various methods for extraction
  - **Classification**: identify keywords
  - **Regression/numerical problems**: identify numbers
  - **Code**: pull out code snippets in triple-backticks

Output Mapping

- Given an answer, map it into a class label or continuous value

- Often map many extracted worlds onto a single class



Zero-shot, Few-shot Prompting / In-context Learning

Zero-shot Prompting

- Solve a task by simply conditioning the models on an instruction describing the task

Instruction | Please classify movie reviews as 'positive' or ‘negative’.

Few-shot Prompting (Brown et al. 2021)

- Provide a few examples of the task together with the instruction

Instruction

```
Please classify movie reviews as 'positive' or ‘negative’.
```

Examples

```
Input: I really don’t like this movie.
Output: negative

Input: This movie is great!
Output: positive
```

LMs are Sensitive to Small Changes in In-context Examples

- Example ordering (Lu et al. 2021)

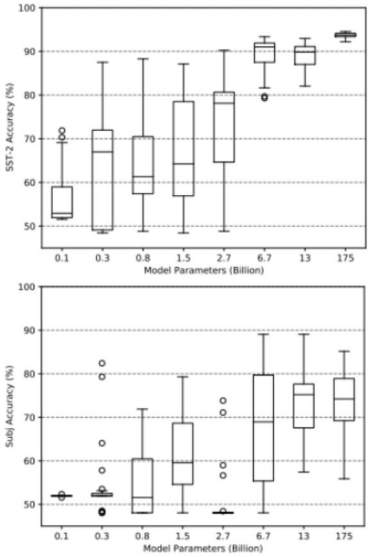
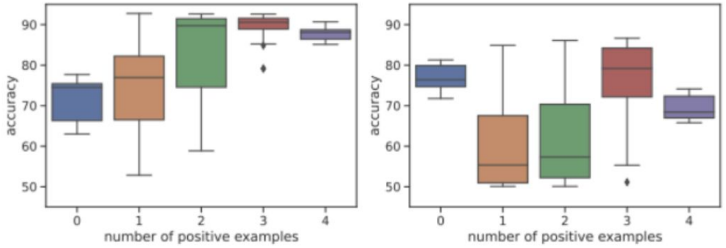


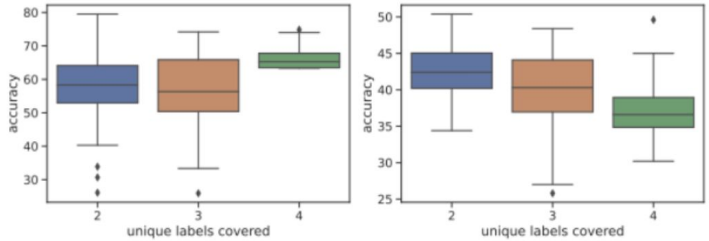
Figure 1: Four-shot performance for 24 different sample orders across different sizes of GPT-family models (GPT-2 and GPT-3) for the SST-2 and Subj datasets.

- Label balance (Zhang et al. 2022)



(a) Amazon (b) SST-2

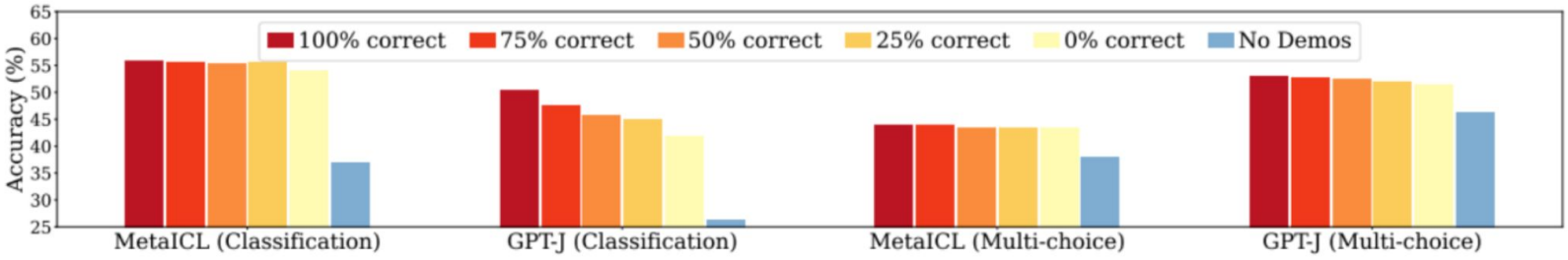
- Label coverage (Zhang et al. 2022)



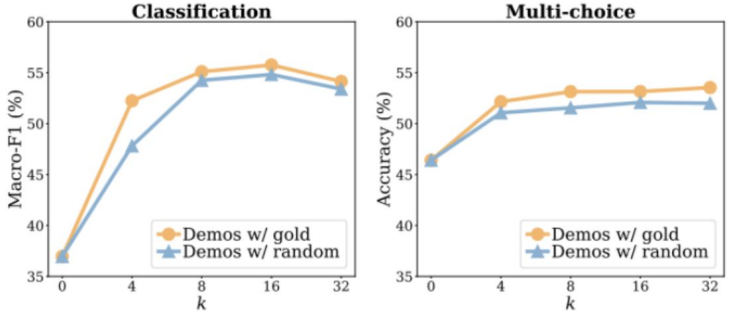
(a) AGNews (b) TREC

But Effects are Sometimes Counter-intuitive (Min et al. 2022)

- Replacing correct labels with random labels sometimes barely hurts accuracy



- More demonstrations can sometimes hurt accuracy



Prompt Engineering Basics

Manual Engineering: Instructions

- Instructions should be clear, concise and easy to understand
- Good examples: https://www.promptingguide.ai/introduction/tips

Less Precise:

Explain the concept prompt engineering. Keep the explanation short, only a few sentences, and don't be too descriptive.

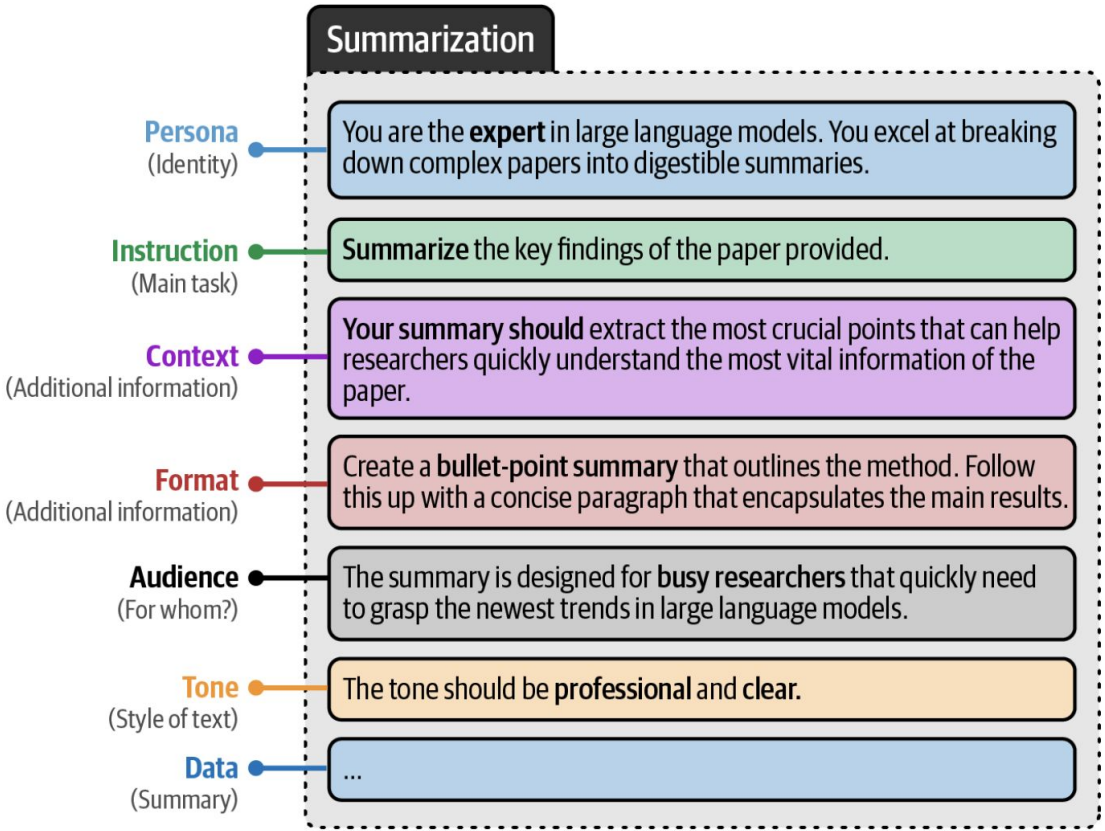
More Precise:

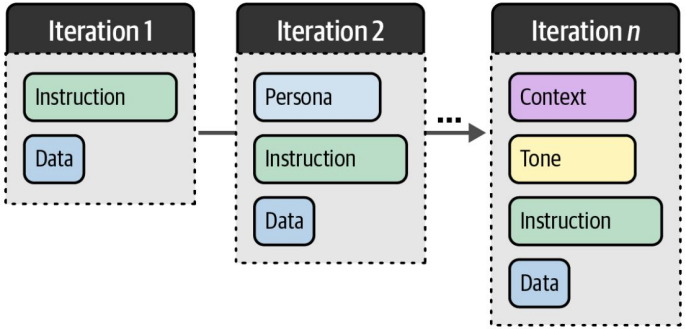
Use 2-3 sentences to explain the concept of prompt engineering to a high school student.

- Similar to humans, but (right now) LMs don’t complain when you’re vague

Prompt Components

- **Persona**: Define the role the LLM should assume. For example, specify “You are an astrophysics expert” to tailor the response to questions on that topic.
- **Instruction**: Clearly outline the task. Ensure it is detailed to minimize ambiguity.
- **Context**: Provide background information to clarify the task’s purpose and reasoning.
- **Format**: Specify the desired output format to ensure consistency, especially in automated systems.
- **Audience**: Identify the intended recipient or level of the output, such as using “ELI5” (Explain Like I’m 5) for simplified explanations.
- **Tone**: Indicate the tone of voice for the response. For formal communication, such as emails to a superior, avoid casual language.
- **Data**: Include any relevant data necessary for completing the task.

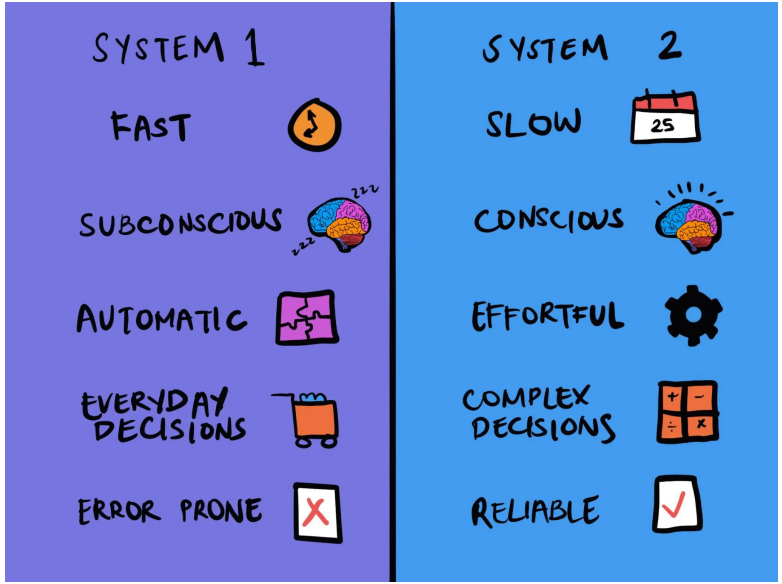

Summarization
Persona
(Identity)
You are the **expert** in large language models. You excel at breaking down complex papers into digestible summaries.
Instruction
(Main task)
Summarize the key findings of the paper provided.
Context
(Additional information)
Your summary should extract the most crucial points that can help researchers quickly understand the most vital information of the paper.
Format
(Additional information)
Create a **bullet-point summary** that outlines the method. Follow this up with a concise paragraph that encapsulates the main results.
Audience
(For whom?)
The summary is designed for **busy researchers** that quickly need to grasp the newest trends in large language models.
Tone
(Style of text)
The tone should be **professional** and **clear.**
Data
(Summary)
...


Iteration 1
Instruction
Data
Iteration 2
Persona
Instruction
Data
Iteration *n*
Context
Tone
Instruction
Data

Advanced Prompting Methods / Reasoning in LMs

What is Reasoning?

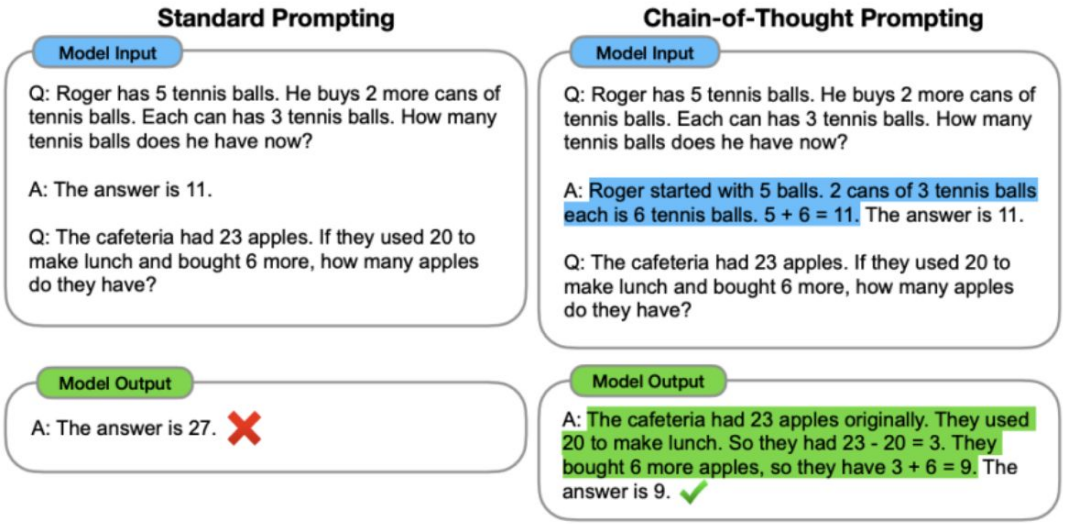
- Using **evidence** and **logic** to arrive at conclusions and make judgment
- Can be
  - Formal: focusing on stric truth values
  - Informal: based on experience and intuition
- How can we design LLMs to mimic reasoning processes and enhance their outputs?



Img: https://thedecisionlab.com/reference-guide/philosophy/system-1-and-system-2-thinking

Chain of Thought (CoT) Prompting (Wei et al. 2022)

- Get the model to explain its reasoning before making an answer



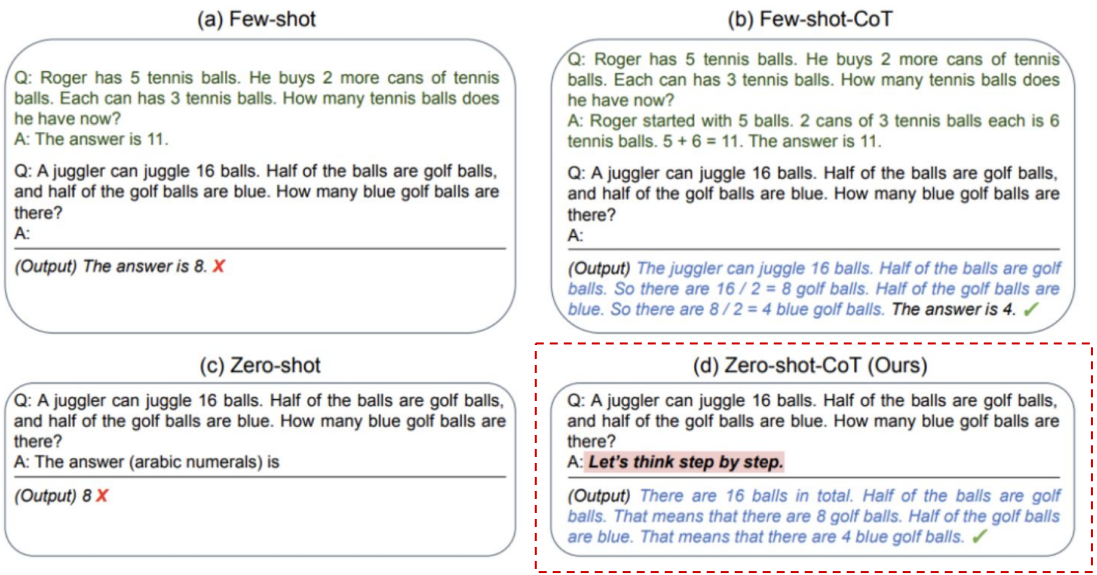
- Provides the model with adaptive computation time

Wei et al. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. https://arxiv.org/abs/2201.11903

Unsupervised Chain-of-thought Prompting (Kojima et al. 2022)

- Just adding a prompt that encourages the model to explain decisions can induce reasoning

- Note: GPT models reason even w/o specific instructions now (probably due to instruction tuning)



Kojima et al., 2022. Large Language Models are Zero-Shot Reasoners. https://arxiv.org/abs/2205.11916

Problem Decomposition

- Solving a complex problem by breaking it down into simpler problem-solving tasks

"Please write a blog about the risks of AI"

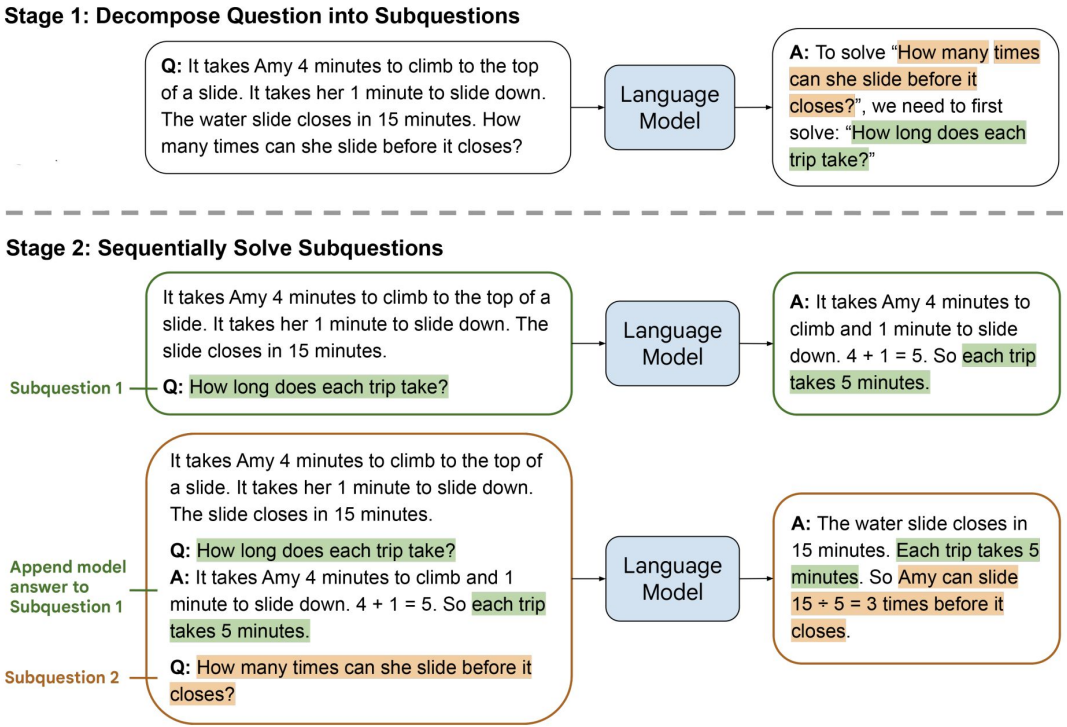
You are a blog writer. Please follow the provided outline below to write a blog about the risks of AI.

- Introduction
  Introduce AI, its relevance, and the importance of understanding its risks for youth.
- Privacy Concerns
  Discuss how AI might compromise personal privacy through interactions online.
- Misinformation
  Explore AI's role in spreading misinformation and influencing young people's decisions.
- Cyberbullying
  Highlight how AI tools can be utilized in cyberbullying and the impact on mental health.
- Tips for Safe AI Use
  Offer guidelines for responsible AI usage and promote critical thinking.
- Conclusion
  Recap main points and encourage proactive engagement with AI ethics.

Img: Xiao and Zhu 2025. Foundations of Large Language Models. https://arxiv.org/abs/2501.09223

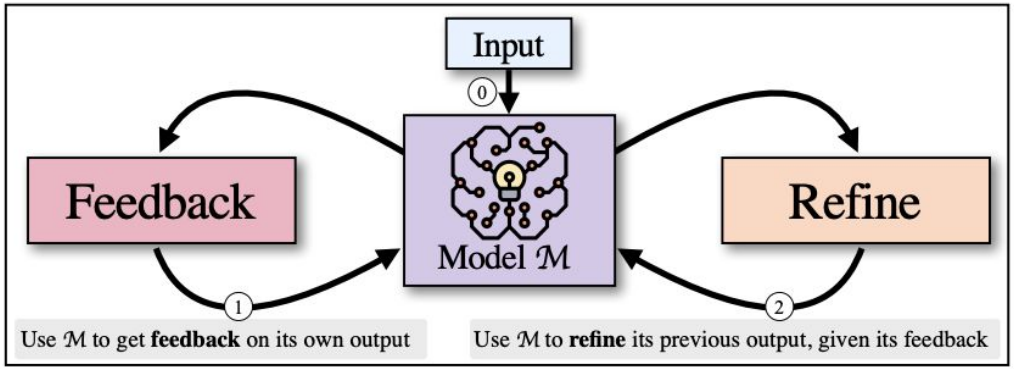
Least-to-Most Prompting (Zhou et al. 2023)

- Sub-problem generation is performed by prompting an LLM with instructions and/or demonstrations.
- Solving each subproblem is facilitated by the answers to previously solved subproblems.



Zhou et al. 2023. LEAST-TO-MOST PROMPTING ENABLES COMPLEX REASONING IN LARGE LANGUAGE MODELS. https://arxiv.org/abs/2205.10625

Self-refinement (Madaan et al., 2024)



- **Prediction**. We use an LLM to produce the initial model output.
- **Feedback** Collection. We obtain feedback on the model output.
- **Refinement**. We use the LLM to refine the model output based on the feedback.

Madaan et al., 2024. Self-Refine: Iterative Refinement with Self-Feedback. https://arxiv.org/abs/2303.17651

Self-refinement (Madaan et al., 2024)

(a) **Dialogue:** $x, \mathbf{y_t}$

```
User: I am interested
in playing Table
tennis.

Response: I'm sure
it's a great way to
socialize, stay active
```

(b) **FEEDBACK** $\mathbf{fb}$

```
Engaging: Provides no
information about table
tennis or how to play it.

User understanding: Lacks
understanding of user's
needs and state of mind.
```

(c) **REFINE** $\mathbf{y_{t+1}}$

```
Response (refined): That's
great to hear (...) ! It's
a fun sport requiring
quick reflexes and good
hand-eye coordination.
Have you played before, or
are you looking to learn?
```

(d) **Code optimization:** $x, \mathbf{y_t}$

```
Generate sum of 1, ..., N
def sum(n):
    res = 0
    for i in range(n+1):
        res += i
    return res
```

(e) **FEEDBACK** $\mathbf{fb}$

```
This code is slow as
it uses brute force.
A better approach is
to use the formula
... (n(n+1))/2.
```

(f) **REFINE** $\mathbf{y_{t+1}}$

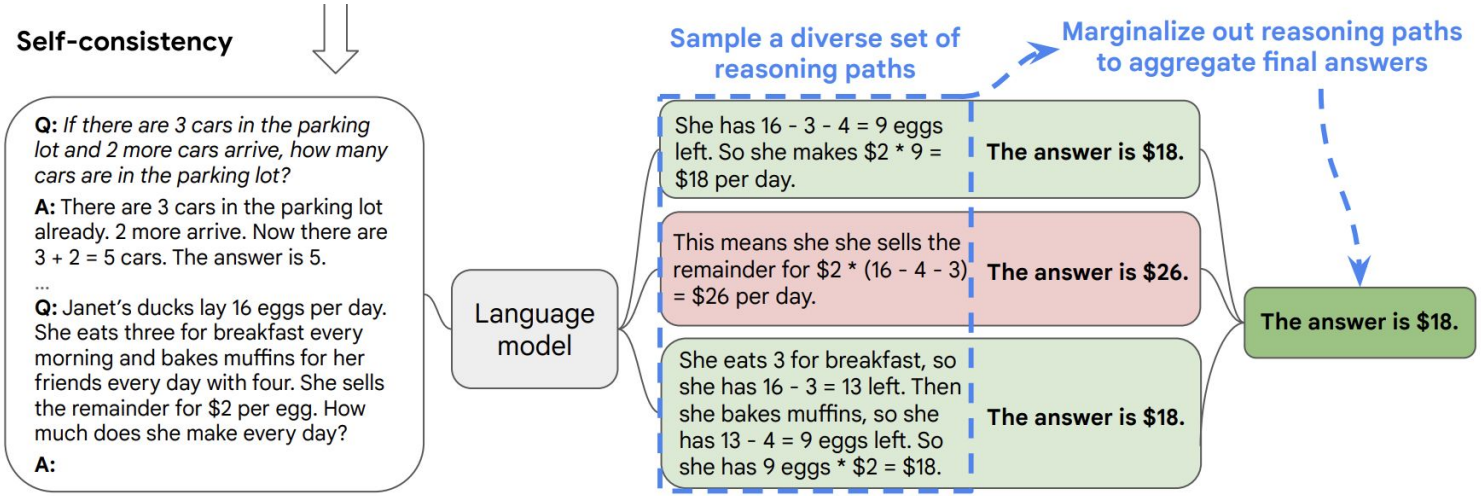
```
Code (refined)

def sum_faster(n):
  return (n*(n+1))//2
```

Madaan et al., 2024. Self-Refine: Iterative Refinement with Self-Feedback. https://arxiv.org/abs/2303.17651

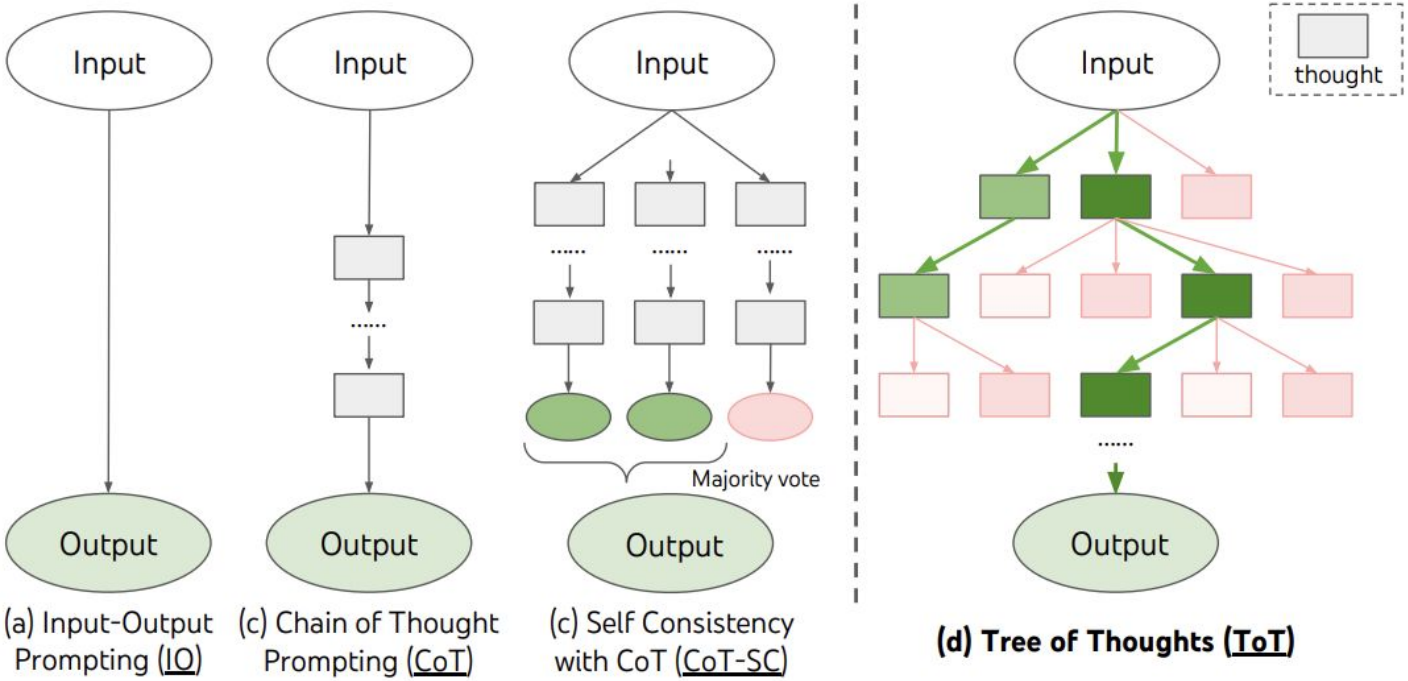
Self-consistency (Wang et al., 2023)

- It first samples a diverse set of reasoning paths instead of only taking the greedy one, and then selects the most consistent answer by marginalizing out the sampled reasoning paths.



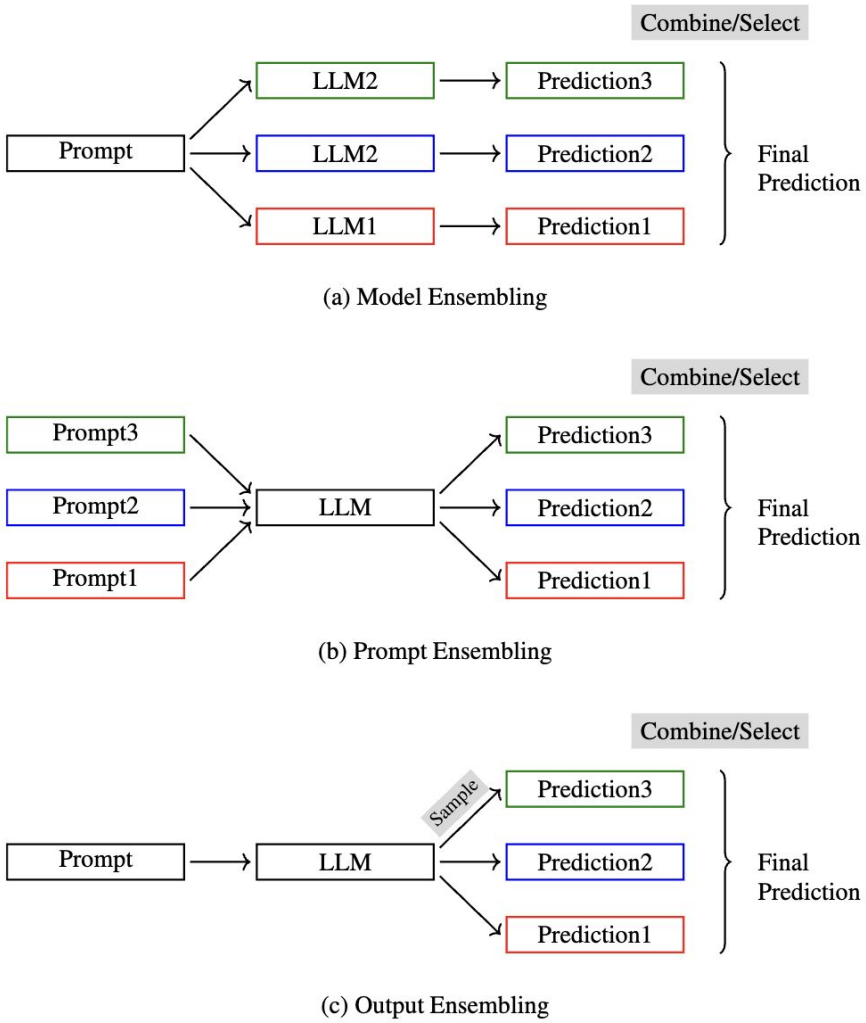
Wang et al., 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. https://arxiv.org/abs/2203.11171

Tree-of-Thought: Exploring Intermediate Steps



Yao et al. 2023 Tree of Thoughts: Deliberate Problem Solving with Large Language Models. https://arxiv.org/abs/2305.10601

Ensembling methods for LLMs



Img: Xiao and Zhu 2025. Foundations of Large Language Models.
https://arxiv.org/abs/2501.09223

Any Questions?